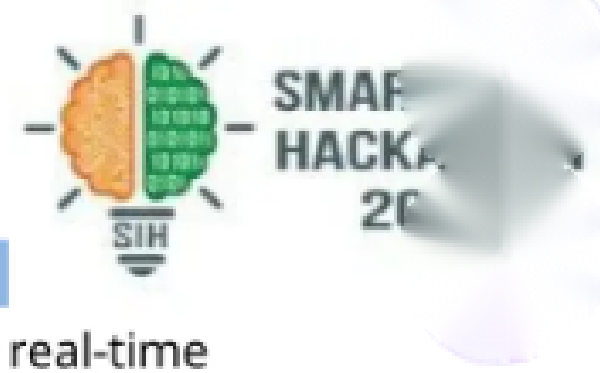


TITLE PAGE

- **Problem Statement ID – SIH25134**
- **Problem Statement Title** - There is a need to design drones and robots that can solve some of the pressing challenges of India such as handling medical emergencies, search and rescue operations, etc
- **Theme - Robotics and Drones**
- **PS Category - Software**
- **Team ID -**
- **Team Name – TEAM STARFLEET**





PROPOSED SOLUTION

- ❖ Detailed explanation: Developing autonomous drones and rugged robots equipped with AI, sensors, and real-time communication can deliver medical aid, locate victims, and support disaster rescue. These systems ensure rapid response, reduce human risk, and enhance India's emergency preparedness capabilities.
- ❖ How it addresses the problem:
 - **Medical Emergency: Drones** deliver essential supplies like, oxygen, and medicines within minutes, bypassing traffic and reaching remote villages.
 - **Search & Rescue Operations: Thermal cameras, AI-based victim detection, and audio sensors** on drones identify survivors under rubble.
 - **Integration & Scalability**: Scalable design allows deployment in both **urban traffic scenarios** and **rural inaccessible regions**.

The solution directly addresses the problem statement by using drones for speed and accessibility, and an integrated digital platform for coordination, making emergency handling and rescue faster, safer, and more reliable.

INNOVATION & UNIQUENESS

1. **AI-powered Victim Detection**: Uses **multi-sensor fusion** (thermal + visual + audio) with lightweight AI models to **detect survivors**.
2. **Payload Customization**: Drones are equipped with **modular payload pods** for medicines depending on the emergency type
3. **Smart Dashboard & Analytics**: Centralized **command dashboard** with live drone/robot feeds, GIS mapping, AI-based triage, and predictive analytics for **decision-making and resource optimization** during emergencies.



TECHNICAL APPROACH

HARDWARE

1. Strong Plastic Frame
2. Brushless DC Motors (BLDC) – 935 KV
3. Propellers – 10-12 inch
4. Electronic Speed Controllers – 15-20 A
5. Flight Controller - Pixhawk
6. LiPo Battery 4s
7. Transmitter (TX) & Receiver (RX) – fs i10
8. GPS Module & Compass
9. Raspberry Pi
10. FPV Camera
11. Thermal Camera
12. Ground Support
13. Servo Motor – Hook/Drop

TECHSTACK USED

? Frontend

Next.js, React, Tailwind CSS

? Backend

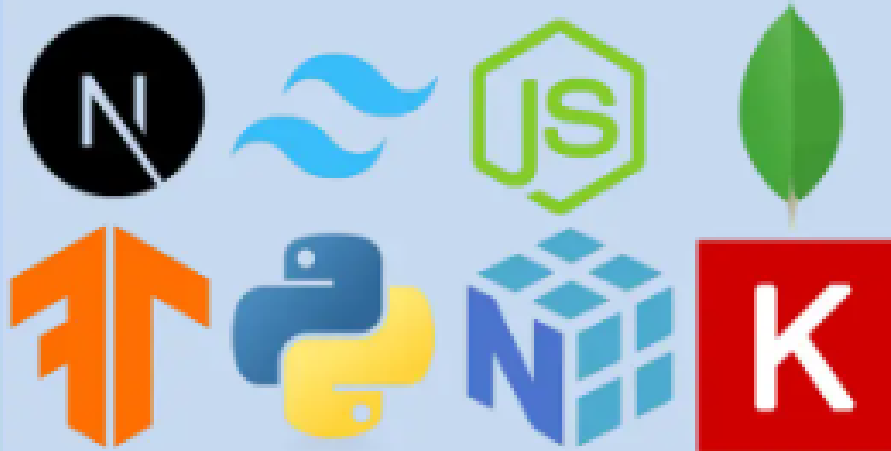
Node.js, express.js, MongoDB Atlas

? Dashboard & AI

Tensorflow Lite, Python, Mission Planner

? Python Libraries

Numpy, math.lib, keras, scikit learn



FLOWCHART

Emergency Triggered
(Manually, sensor alert)



Command Dashboard
(AI based request & allocation)



Drone Dispatch
(Meds & Surveillance)

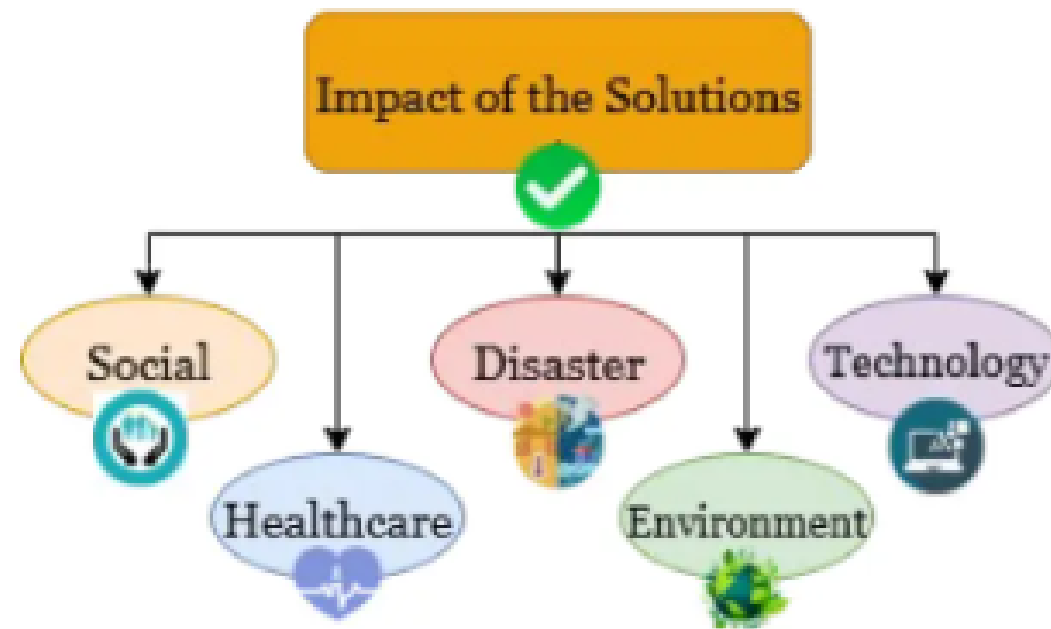


Live Monitorin(g)
(AI Victim Detection)



Data Sync & Analytics
(GIS mapping & Response time logs)

IMPACT AND BENEFITS



IMPACTS OF THE SOLUTIONS

Social

- Lives saved by reducing response times in accidents, and disasters.
- Helps people in remote villages, hilly areas, or flood-prone regions where ambulances struggle.

Healthcare

- **First response within minutes** – AEDs, medicines, and oxygen delivered much faster than conventional ambulances.
- Provides **contactless services** during pandemics to reduce infection spread.

Disaster Management Impact

- Drones provide **real-time mapping** of disaster-hit zones.
- Robots support **search & rescue** in collapsed or unsafe areas.

Environmental Impact

- Monitors **forest fires, floods, landslides & pollution** levels.
- Supports **precision agriculture**, reducing resource wastage.

Technological Impact

- Advances **AI, IoT, robotics & drone technology**.
- Opens opportunities for **startups & MSMEs** in robotics & automation

BENEFITS:

- Provides **real-time mapping and surveillance** of disaster-affected areas.
- Supports **precision agriculture**, cutting down water, fertilizer, and pesticide wastage.
- Reduces dependency on **imported technologies** by developing indigenous solutions.
- Helps in **wildlife monitoring, habitat protection, and anti-poaching efforts**.
- Improves overall **emergency response speed**, directly saving lives.
- Promotes **sustainability and efficient resource management**



FEASIBILITY AND VIABILITY



❖ Feasibility of the idea

- **Affordable & Scalable** - Low-cost drone and robotic solutions can be developed using indigenous tech
- **Wide Applications** - One design can be adapted for healthcare, disaster relief, agriculture, and environment.
- **Supportive Ecosystem** - Govt. initiatives like Drone Shakti and Atmanirbhar Bharat encourage adoption.
- **Existing Tech Base** - India already has drone policies, AI & IoT infrastructure to build on.

❖ Viability

- **Market Demand:** High need in healthcare, disaster management, agriculture, and environmental monitoring
- **Government Support:** Backed by policies Atmanirbhar Bharat.
- **Commercialization:** Viable as a startup model with applications in public and private sectors.

❖ Implementation

- Research & identify key use-cases
- Design modular prototype (drones/robots)
- Develop hardware + software integration
- Test in lab & controlled field environments
- Optimize battery, payload & autonomy

❖ Potential challenges and risks

Technical Failures:
Navigation errors, sensor malfunction, software glitches.

Battery & Payload:
Limitations, limited flight time, carrying capacity.

Safety Risks:
Accidents in populated areas.

Environmental Factors:
Weather affecting performance.

Regulatory Compliance:
Permissions, airspace restrictions.

High Initial Costs:
Prototype medical/development, testing.

Data Security & Privacy:
Surveillance data.

Maintenance & Durability:
Wear & tear, repair, repair in remote areas.

Adoption Resistance:
Hesitation from authorities/users.

❑ MARKET SIZE

Category	Converted Value (INR) (Approx. at 1 USD = ₹83)
Drone Market	6,90,000 Crore
Commercial Market	1,43,000 Crore
Industrial Robotics	3,15,000 Crore
Collaborative Robots	46,000 Crore
Advanced Robotics	4,42,000 Crore

ARTICLES & RESEARCH PAPER

1. Application of drone in disaster management: A scoping review – [Article](#)
2. Telecommunication Engineering Centre Department of Telecommunications Ministry of Communications Government of India – [Use cases of drones in disaster management](#)
3. National Institute of Health – [Steps to build a DIY low-cost fixed-wing drone](#)
4. IIT Madras – [Drone Design and fabrication](#)